

Credibility assessment of explicit FEA for deployment force in a transcatheter delivery system: ASME V&V40 application across two models and two mechanisms

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Abstract

Purpose: To assess the credibility of two explicit finite element analysis (FEA) models used to estimate deployment forces in a transcatheter delivery system for a self-expanding peripheral stent, following the ASME V&V40 risk-informed framework. **Scope:** Model 1 simulates lesion-crossing dynamics to compute peak advancement force, force–displacement profile, and local wall strain. Model 2 simulates sheath retraction and release to compute transient force spikes and steady retraction force while characterizing stick–slip and energy dissipation. Two clinical mechanisms motivate the assessment: lesion crossing (Mechanism A) and sheath retraction for release (Mechanism B). **Approach:** We define a single context of use (CoU) aimed at informing bench-equivalent deployment force ranges and spike amplitudes for design selection and test planning. We adopt a private 0–12 gradation scale with a per-pair acceptance threshold of 8. Credibility factors considered are numerical code verification, numerical solver error, output comparison, relevance of validation activities to the CoU, software quality assurance, and test samples. Use error is addressed qualitatively. Methods include explicit-dynamics verification benchmarks (e.g., Taylor bar, contact patch), solver error studies (time step refinement, mass scaling limits, contact stabilization), instrumented bench correlation using silicone arterial phantoms with embedded calcifications, mapping of validation coverage to clinical use, and a controlled model software pipeline. **Results:** For Model 1 under Mechanism A, agreement with bench force–displacement data was strong (RMSE 0.58 N; R² 0.94; peak 7.1 ± 0.7 N vs 7.4 ± 0.6 N), solver errors were limited ($\leq 3\%$ change between final refinement steps; kinetic energy fraction $< 7\%$), and verification residuals were $\leq 1.5\%$. Scores exceeded threshold in all six factors (11/12, 10/12, 9/12, 10/12, 12/12, 10/12). For Model 2 under Mechanism B, the spike and steady retraction forces matched bench tests (RMSE 0.87 N; spike lag < 6 ms; spike 12.6 ± 1.3 N vs 13.2 ± 1.0 N), with solver sensitivity somewhat higher due to contact regularization but within limits ($\leq 4\%$ change across final refinements). Scores exceeded threshold in all six factors (11/12, 9/12, 10/12, 10/12, 12/12, 9/12). Cross-mechanism applications (Model 1 $\times$ Mechanism B; Model 2 $\times$ Mechanism A) did not reach the acceptance threshold due to limited validation mapping and test coverage for those misaligned uses, despite strong verification and software rigor. **Conclusions:** Within the declared CoU, the paired uses Model 1 $\times$ Mechanism A and Model 2 $\times$ Mechanism B satisfy the threshold for risk-informed credibility and can be used to inform design and test planning for deployment forces and spikes. The out-of-scope pairs are not accepted for decision support. We discuss limitations, including phantom generalizability and contact modeling assumptions, and outline plans for extended validation across tortuosity and calcification morphologies, and for probabilistic sensitivity analysis.

Keywords: ASME V&V40, explicit finite element, transcatheter deployment

1. Abstract

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adopt a private 0–12 gradation scale with a per-pair acceptance threshold of 8. Credibility factors considered are numerical code verification, numerical solver error, output comparison, relevance of validation activities to the CoU, software quality assurance, and test samples. Use error is addressed qualitatively.

Methods include explicit-dynamics verification benchmarks (e.g., Taylor bar, contact patch), solver error studies (time step refinement, mass scaling limits, contact stabilization), instrumented bench correlation using silicone arterial phantoms with embedded calcifications, mapping of validation coverage to clinical use, and a controlled model software pipeline.

Results: For Model 1 under Mechanism A, agreement with bench force–displacement data was strong (RMSE 0.58 N; R² 0.94; peak 7.1 ± 0.7 N vs 7.4 ± 0.6 N), solver errors were limited ($\leq 3\%$ change between final refinement steps; kinetic energy fraction $< 7\%$), and verification residuals were $\leq 1.5\%$.

Scores exceeded threshold in all six factors (11/12, 10/12, 9/12, 10/12, 12/12, 10/12). For Model 2 under Mechanism B, the spike and steady retraction forces matched bench tests (RMSE 0.87 N; spike lag <6 ms; spike 12.6 ± 1.3 N vs 13.2 ± 1.0 N), with solver sensitivity somewhat higher due to contact regularization but within limits ($\leq 4\%$ change across final refinements). Scores exceeded threshold in all six factors (11/12, 9/12, 10/12, 10/12, 12/12, 9/12). Cross-mechanism applications (Model 1 $\times$ Mechanism B; Model 2 $\times$ Mechanism A) did not reach the acceptance threshold due to limited validation mapping and test coverage for those misaligned uses, despite strong verification and software rigor.

Conclusions: Within the declared CoU, the paired uses Model 1 $\times$ Mechanism A and Model 2 $\times$ Mechanism B satisfy the threshold for risk-informed credibility and can be used to inform design and test planning for deployment forces and spikes. The out-of-scope pairs are not accepted for decision support. We discuss limitations, including phantom generalizability and contact modeling assumptions, and outline plans for extended validation across tortuosity and calcification morphologies, and for probabilistic sensitivity analysis.

2. Background and objective

Self-expanding stents delivered through peripheral arteries are deployed using long, flexible catheter systems. Operators must traverse stenotic segments with sufficient axial force to advance the delivery sheath (lesion crossing) and then retract the outer sheath to release the stent (sheath retraction). Both maneuvers depend on frictional contact and local compliance of the vessel–plaque complex, and both carry risks if forces are excessive or rapidly changing. Computational modeling of these maneuvers can support design decisions and bench planning, provided the models achieve a level of credibility commensurate with their intended use.

The present work assembles a structured credibility argument for two explicit finite element models. The first targets the advancement phase and computes peak push force, the force–displacement response during crossing, and localized wall strain where the sheath bears on the lesion. The second targets sheath withdrawal, quantifying transient spikes and steady force due to stick–slip behavior at the device–sheath–vessel interfaces. We apply a risk-informed process patterned after ASME V&V40, including verification activities, solver error characterization, comparison to targeted bench experiments, mapping of validation evidence to the context of use (CoU), control of the modeling software pipeline, and disciplined handling of experimental test samples.

Our objective is to determine, for each model under each mechanism of interest, whether the available evidence—on a private 0–12 gradation scale—exceeds a prespecified threshold for acceptability. The analysis is designed not only to yield a pass–fail judgment but also to identify where confidence is strong and where it is insufficient, guiding subsequent development.

Relevance of validation activities to the COU — gradation.

- a. Minimal resemblance between tests and intended maneuvers; limited traceability of inputs and outputs.
- b. Partial alignment on either maneuvers or ranges of inputs; qualitative linkage between test metrics and COU decisions.
- c. Strong alignment on maneuvers and inputs; quantitative mapping of test metrics to COU outputs; minor gaps in boundary conditions.
- d. Direct operational match on maneuvers, inputs, and outputs; quantitative coverage map with statistical confidence on extrapolation limits.

3. Device and clinical motivation

The device is a 6–8 Fr transcatheter delivery system for a laser-cut, self-expanding nitinol stent indicated for femoropopliteal lesions. The delivery system comprises an inner shaft with the constrained stent, an outer polymeric sheath with a lubricious inner liner, and a proximal handle with push and retract controls. During the case, the operator advances the sheath–stent assembly across stenotic segments that may be fibrocalcific, and then releases the stent by proximally withdrawing the outer sheath while stabilizing the inner shaft. The magnitudes and temporal patterns of forces during these maneuvers influence controllability and safety. Excessive advancement force risks forward jump upon sudden loss of resistance, potentially causing geographic miss or vessel wall trauma. During release, sudden spikes in retraction force can induce axial motion of the delivery system and transient wall loading.

Clinical mishaps associated with these mechanisms often occur in zones of mixed plaque composition and high curvature. Preclinical bench tests typically use uniform silicone vessels or bovine arteries with simplified lesions, which may not always replicate contact mechanics at fibrotic or calcified plaques. A computational framework, if validated against instrumented experiments that better represent key clinical features, can help delineate safe design envelopes before expensive testing on more complex surrogate or cadaveric platforms.

4. Context of use (CoU) and model risk level under ASME V&V40

We define the CoU as follows: to inform design selection and bench test planning by predicting ranges of advancement force during crossing of stenotic, partially calcified segments, and to predict transient and steady force during sheath retraction for stent release, using models calibrated only to generic material data and evaluated for agreement with targeted bench tests. The intended decisions include down-selection among candidate sheath materials and lubricious liner combinations, and specification of bench test conditions to stress the system

within safe force envelopes. The outputs are the force–displacement curve and peak force for crossing and the transient spike amplitude, spike timing, and steady force for retraction. Predictive use is limited to straight or mildly curved vessels (curvature radius ≥ 60 mm), 40–80% diameter stenoses with partial circumferential calcification (0–20% volume fraction by microbeads in phantom), and physiological saline at 37 °C.

Following the risk-informed categorization of ASME V&V40, we judge the model influence on decision-making to be moderate: modeling outputs are combined with bench data rather than replacing them, and the consequences of an erroneous prediction include suboptimal bench stress selection but not direct clinical harm. The model consequence is indirect, and the decision consequence is moderate since it could lead to retesting or redesign. Accordingly, a medium to high level of credibility is sought, particularly for the alignment between bench validation and the declared CoU. We introduce a private, integer 0–12 gradation scale for each factor assessed, with 0 meaning none and 12 meaning excellent. The acceptance threshold for each model $\times$ mechanism pair is set at 8. Scores are paired with mechanistic scope and are not pooled across mechanisms; this ensures that a high score under one maneuver does not obscure a gap under another.

We delineate two mechanisms aligned to clinical maneuvers: Mechanism A (lesion crossing) and Mechanism B (sheath retraction for release). Each model is assessed under both mechanisms to document where evidence is sufficient and where it remains insufficient for decision support within the CoU.

Output comparison — gradation.

- a. Anecdotal or qualitative alignment to non-instrumented observations; no uncertainty quantification.
- b. Single-condition quantitative comparison with limited repeatability; basic error metrics without propagation of measurement uncertainty.
- c. Multi-condition quantitative comparison with replication and uncertainty; pre-specified acceptance criteria; minor mismatches addressed by sensitivity studies.
- d. Blind, multi-condition prediction with cross-validation; full accounting of measurement and model uncertainty; pre-registered acceptance criteria met.

5. Computational framework and quantities of interest

Both models are implemented using an explicit dynamics finite element code executed within a containerized environment on dual-socket x86 Linux workstations (2×24 -core CPUs, 256 GB RAM). The solvers employ central difference time integration with single-precision contact kinematics and double-precision global reductions. Contact is treated with a penalty-based surface-to-surface formulation with adaptive penalty scaling. Kinematic constraints emulate the handle actuation and guidewire support. The inner shaft and constrained

stent are represented by reduced-order beam and shell models, respectively, with effective stiffness matching measured load–deflection curves. The outer sheath is modeled as a multi-layer composite: a Pebax jacket (shore D 55–72) and a PTFE liner (50–80 μm). The artery is modeled as a three-layer hyperelastic tube using a Holzapfel–Gasser–Ogden formulation with parameters fit to published femoropopliteal data; the lesion is a circumferentially inhomogeneous ring with regions of high stiffness representing calcified inclusions, inserted using level-set–defined material assignments. Fluid effects are not modeled explicitly; the lumen contains saline at 37 °C and acts to lubricate contacts via reduced friction coefficients calibrated from pin-on-disk tests on PTFE–Pebax and PTFE–silicone pairs.

Quantities of interest are distinct for each mechanism. For lesion crossing, we compute: (i) axial force at the proximal shaft measured via a virtual load cell, (ii) advancement displacement and velocity of the sheath leading edge, (iii) peak force and its displacement location, and (iv) maximum principal strain within the vessel wall at the lesion contact zone. For sheath retraction, we compute: (i) proximal retraction force at the handle, (ii) temporal evolution of force spikes during stick–slip release of the stent–sheath interface, (iii) time lag between the onset of motion and force peak, (iv) energy dissipated through friction at the device–sheath and sheath–vessel interfaces, and (v) post-spike steady retraction force. Outputs are post-processed at 1 kHz equivalent sampling with a 50 Hz low-pass Butterworth filter applied to reduce high-frequency numerical chatter without obscuring spike characteristics.

Model inputs include geometric tolerances (± 0.02 mm on sheath inner diameter, ± 0.03 mm on outer diameter), material moduli (Pebax 450–700 MPa, PTFE 400–600 MPa in tension; nonlinearity accounted via Ramberg–Osgood fits), friction coefficients (PTFE–silicone 0.05–0.12; PTFE–nitinol 0.08–0.15; Pebax–lesion 0.20–0.35), and lesion stiffness contrasts (calcified inclusion up to 20 $\times$ baseline medial stiffness). Material damping is represented by Rayleigh coefficients tuned to match ring-down tests of isolated components, yielding modal Q-factors in the range 20–40 for frequencies up to 500 Hz. All inputs are versioned and archived alongside meshes and solver decks.

The benchmarks and validation experiments drive calibration only of generic, component-level parameters (e.g., friction) but not force–displacement or spike outputs; those are treated as validation-only metrics. We enforce a separation of calibration and validation to reduce overfitting risk.

Numerical solver error — gradation.

- a. Single nominal time step and mesh with no sensitivity study; mass scaling without justification.
- b. Limited refinement study on time step or mesh; basic kinetic-to-internal energy checks; qualitative mass scaling limits.
- c. Systematic refinement on time step and contact regularization with quantitative convergence of outputs; mass scaling bounded by energy and inertia metrics.



Figure 1: Schematic of the computational workflow: inputs (geometry, materials, contact), solver (explicit dynamics, contact), outputs (force, displacement, spike), and comparison to bench tests with data management and traceability hooks.

- d. Combined spatial–temporal convergence with extrapolation of outputs; multi-parameter response surfaces; uncertainty reported for solver-induced error.

6. Model 1 — Lesion-Crossing Dynamics: geometry, materials, contacts, and loads

Model 1 constructs a straight arterial segment of 150 mm length with an inner diameter of 5.5–6.5 mm and wall thickness of 0.9–1.1 mm, discretized with 8-node hexahedra in three layers through the wall (typical element size 0.4–0.6 mm). A 15–25 mm long lesion is embedded as an annular constriction yielding 40–80% area stenosis, with an eccentric calcified arc of 60–120° represented by a stiffened subregion (elastic modulus 2–5 GPa). The delivery sheath advances from proximal to distal across the lesion at controlled velocities of 5–25 mm/s. The inner shaft is tethered axially at the handle location but free to rotate; the sheath is pushed by a prescribed displacement at the proximal end via a rigid connector that collects axial force outputs. Contacts include: sheath liner to inner shaft (low friction), sheath outer to vessel lumen (moderate friction modulated by saline lubrication), and sheath leading edge to lesion shoulder (high local curvature contact). The plaque–sheath contact is the dominant contributor to peak force.

Sheath layers are modeled with 4-node reduced-integration shells (0.2–0.4 mm target edge length) using orthotropic material properties to capture layer anisotropy. The PTFE liner is assigned a very low shear modulus to mimic its lubricity, while the Pebax jacket carries most of the bending stiffness. A geometric imperfection (0.05–0.1 mm ovalization) is imposed on the sheath cross-section to reflect manufacturing tolerances and to activate realistic contact patch evolution at the lesion shoulder. Loading consists of a ramped push displacement over 1–3 s segments, with median advancement rate 10 mm/s. The boundary conditions fix the proximal guidewire and limit distal vessel translation to emulate fixture grips in the bench setup.

Mesh convergence for wall strain and contact pressures was confirmed by comparing 0.4 mm and 0.25 mm element sizes in the lesion region, which changed peak strain by 2.6% and contact maximum pressure by 4.1% while increasing cost by 2.7×. These checks guided the selection of the baseline mesh for solver error studies and validation comparisons. The force–displacement output is dominated by the formation and release of a contact

ridge at the proximal lesion shoulder, and the selected meshing captures the ridge width (1.3–1.8 mm) within 0.2 mm when benchmarked against high-resolution scans of indented silicone plaques.

Test samples — gradation.

- a. Untargeted or convenience specimens; uncontrolled conditioning; unreported fixtures.
- b. Basic targeting of geometry with limited metrology; some conditioning control; fixtures documented at a high level.
- c. Targeted morphologies with dimensional verification; controlled environment and preconditioning; fixtures measured for alignment and compliance.
- d. Statistically representative morphologies with traceable metrology; full environmental control and aging; fixture compliance and alignment uncertainties quantified and propagated.

7. Model 2 — Sheath Retraction and Release: geometry, materials, contacts, and loads

Model 2 reutilizes the vessel and sheath representations but includes a fully detailed, crimped nitinol stent with realistic strut–sheath contact. The stent is modeled with shell elements (0.05–0.1 mm edge length) and a superelastic constitutive law calibrated to tension–compression–torsion tests at 37 °C. The crimped outer diameter is 5.5–6.0 mm inside the sheath lumen. The retraction maneuver is represented by prescribing proximal sheath displacement at 10–50 mm/s while restraining the inner shaft axially and allowing small rotations. The stent begins to expand locally as the sheath passes over its proximal edge, leading to stick–slip events as strut apices detach from the liner. The device–sheath–vessel triad creates complex, transient contact patterns and energy dissipation peaks that align with observed force spikes in bench tests.

The dominant model outputs are the instantaneous retraction force at the handle, the amplitude and duration of force spikes, the time of the first spike relative to commanded motion, and the post-spike steady force. The force spikes arise from the sudden release of stored elastic energy in the stent and the transition

of contact states from static to kinetic friction. Penalty contact parameters are tuned to ensure that interpenetration stays under 1% of local element size during spikes, avoiding unphysical force overshoots. Contact damping is used sparingly (nondimensional coefficient 0.05–0.2) to suppress chatter without delaying spike onset beyond the experimental distribution. Mass scaling is constrained such that the ratio of added to physical mass within stent and sheath subdomains remains below 1.08 during retraction.

Vessel curvature is held mild (radius ≥ 60 mm) to confine the CoU to near-straight segments, simplifying the mapping between bench and model. Friction coefficients are scenario-specific: PTFE–nitinol 0.08–0.15 for dry and wetted conditions; PTFE–silicone 0.05–0.12 for the lumen. The sheath jacket–lesion pair remains influential for local wall loading but contributes modestly to retraction spikes compared to the stent–liner interface. Model 2 is therefore more sensitive to liner surface properties and stent crimp uniformity than Model 1.

Numerical code verification — gradation.

- a. No documented benchmarks; reliance on solver vendor claims.
- b. Limited internal checks on element formulation and contact; partial benchmarks without quantitative pass–fail.
- c. Standard benchmarks for dynamics and contact with quantitative tolerances; automated regression on key kernels.
- d. Extensive suite including challenging contact and high-rate cases; tight tolerances; continuous integration with fail-safe triggers.

8. Mechanism A and Mechanism B: boundary conditions and test-representative kinematics

Mechanism A imposes a controlled proximal push on the sheath at velocities representative of clinical advancement—median 10 mm/s, range 5–25 mm/s—while the inner shaft is guided by a low-compliance fixture that constrains axial motion at the handle and permits limited rotations. The distal vessel is gripped by compliant clamps that reproduce the axial stiffness measured in bench fixtures (45–60 N/mm). The guidewire is modeled as a low-friction constraint within the inner shaft lumen, producing minimal additional axial resistance. The initial state sets the sheath just proximal to the lesion shoulder, and the model computes the rising force as the sheath edge engages the stenosis.

Mechanism B applies proximal retraction at velocities of 10–50 mm/s, with the inner shaft axially locked to emulate operator stabilization. The initial state sets the stent crimped under the sheath at target oversizing of 1–2 mm relative to vessel inner diameter. The boundary conditions reproduce the handle compliance (0.03–0.05 mm/N) measured on physical devices to ensure that any elastic energy storage in the handle–shaft path

is captured. The vessel ends are supported as in Mechanism A. The retraction protocol used in bench tests—three repeated pulls per specimen with 30 s rest—has been encoded in the simulation inputs to promote comparable thermal and mechanical histories.

Both mechanisms share a 37 °C saline environment, uniform lubrication assumptions on internal contacts, and controlled initial alignment to within 0.2°. The clinical range of velocities was derived from fluoroscopy-based motion capture in a small set of porcine procedures and corroborated by operator interviews for peripheral interventions. We refrained from including highly tortuous vessel geometries in this assessment to prevent confounding by bending-induced contact states; when curvature is important for a design, a separate CoU will be defined.

Relevance of validation activities to the COU — gradation.

- a. Boundary conditions differ substantially from the clinical maneuvers; inputs not matched to expected ranges.
- b. Partial match in kinematics or loads; boundary conditions approximated; limited justification for differences.
- c. Kinematics, loads, and fixture compliances matched within measured tolerances; differences justified and shown not to affect outputs materially.
- d. Boundary conditions replicated end-to-end; measured distributions used directly; uncertainty in replication propagated to outputs.

9. Numerical code verification (clear): benchmarks, regression tests, and 0–12 scoring

We executed a suite of verification problems targeting element integration, contact enforcement, and high-rate dynamics. For kinematics and energy conservation, a 3D Taylor bar impact test at 200 m/s was performed and compared to semi-analytic solutions and peer benchmark data, yielding a normalized interface position error of 1.3% at 80 μ s and total energy drift under 0.6% over the simulated window. For contact enforcement, a patch test with a rigid spherical indenter pressing on a compliant shell was performed with frictionless contact; nodal reaction force matched the analytical Hertzian solution within 1.4% up to 20% of shell thickness indentation. For frictional sliding, a ring-on-rod sliding test was used; steady sliding force converged to within 1.8% of the analytical value for $\mu = 0.1$ across a range of penalty parameters; stick–slip transients showed no overshoot beyond 3% relative to a high-fidelity reference run with very small time step and tight contact tolerances.

To verify robustness across code changes, the models and kernels run in a continuous integration pipeline: nightly regression consumes 42 short tests covering element stiffness recovery, contact interpenetration limits, and energy accounting in explicit stepping. Over 18 months, no persistent deviation exceeded preset tolerances; two transient deviations were traced to a vendor patch and resolved within a week. We retained

fixed versions of the solver (2025.1) and friction submodule (v3.4.2) for all validation runs in this paper to ensure internal reproducibility. Together, these checks form the foundation for trusting numerical kernels before applying them to the device scenarios.

Numerical code verification — gradation.

- a. Vendor documentation only; no internal checks.
- b. A few internal checks with qualitative expectations; no regression.
- c. Quantified benchmark comparisons with tolerances; automated regression with alerting.
- d. Expanded benchmarks covering nonlinear contact and dynamics; tight tolerances and traceable change control.

10. Numerical solver error (clear): mass scaling, time step, and contact stabilization studies with 0–12 scoring

Solver-induced error was studied by varying the global time step target and local contact regularization while monitoring key outputs. For lesion crossing (Model 1), we compared three nominal time steps: 0.5 μ s, 0.25 μ s, and 0.1 μ s. The force–displacement curve and peak force changed by 5.2% (0.5 to 0.25 μ s) and 2.7% (0.25 to 0.1 μ s). The kinetic-to-internal energy ratio during the peak remained below 0.07 for all steps. For sheath retraction (Model 2), the same sequence of time steps yielded changes in spike amplitude of 6.1% and 3.9%, respectively, with spike timing advancing by up to 4.8 ms at the coarsest step and stabilizing within 1.9 ms at 0.1 μ s. We imposed a mass scaling cap such that the added mass fraction in the stent and sheath domains did not exceed 0.08; across the refinement series, the actual added mass fraction was 0.05–0.07. Force spike amplitude was insensitive to mass scaling within this band, changing by 1.8% when doubling the cap from 0.04 to 0.08 in a controlled run.

Contact stabilization parameters were explored by varying the nondimensional damping from 0.05 to 0.2. For crossing, peak force changed by 1.1% across this range, whereas for retraction, spike amplitude changed by 2.6% and spike timing by 2.1 ms. We also tightened and loosened the contact penalty by $\pm 25\%$; resulting interpenetration stayed under 1% of local element size, with peak force shifts of 1.9% (crossing) and 3.3% (retraction). These results supported the selection of 0.1–0.25 μ s time steps and moderate contact damping.

Spatial discretization effects on force outputs were modest once a lesion-region mesh spacing of 0.25–0.4 mm was achieved, as reported earlier. In combination, the temporal and contact parameter studies show that numerical errors, while non-negligible in transient spikes, are bounded at a level that supports decision-making in the declared CoU. We report the residual solver error as a component of the credibility score for each model–mechanism pair.

Numerical solver error — gradation.

- a. Single setting; no error quantification.
- b. One-parameter sensitivity with qualitative statements; no error bounds.
- c. Systematic refinement with quantitative changes reported; bounded mass scaling; energy ratios monitored.
- d. Multi-parameter convergence with extrapolated error bands; solver-induced uncertainty propagated to outputs.

11. Output comparison (clear): bench test correlation for deployment force and spike amplitude with 0–12 scoring

Validation experiments were designed to replicate the two clinical maneuvers. For crossing, we fabricated twelve silicone arterial phantoms (six nominal inner diameters 5.5–6.5 mm; wall thickness 1.0 mm) with 15–25 mm lesions at 40–80% area stenosis. Calcification was represented by embedding CaCO₃-loaded polyurethane microbead slurries in 60–120° arcs at 0%, 10%, and 20% volumetric fractions. High-resolution CT verified lesion geometry and bead distribution; average diameter reduction was within $\pm 2.3\%$ of targets. The delivery system was mounted on a linear stage with 0.01 mm resolution and advanced at 5, 10, and 20 mm/s through each phantom. A 50 N load cell (0.05 N resolution, $\pm 0.25\%$ accuracy) captured axial forces at 1 kHz. Each condition was repeated three times per specimen with 60 s rest. The resulting force–displacement curves exhibited a characteristic rise at the proximal shoulder, a plateau or second rise through the lesion body, and a decline past the distal shoulder.

For sheath retraction, ten tests were performed across three lots of devices in the same phantoms (where compatible with stent sizes). Retraction velocities were 10, 20, and 40 mm/s, controlled by the linear stage in the proximal direction while the inner shaft was held fixed. The same load cell captured force spikes and steady forces. High-speed video (1000 fps) synchronized with force traces established spike timing relative to motion. The phantoms were filled with 37 °C saline and conditioned by cycling three trial crossings before recorded runs.

Model 1 predictions for crossing matched peak advancement forces with RMSE 0.58 N across the full matrix and had $R^2 = 0.94$ for force–displacement curves after time-alignment by 5–20 ms to account for stage start-up transients. The mean peak from simulations was 7.1 ± 0.7 N versus 7.4 ± 0.6 N measured. The dependence on stenosis severity and calcification fraction was captured, with a slope difference of 0.06 N per 10% stenosis compared to 0.08 N in experiments. Model 2 predictions for retraction reproduced spike amplitudes with RMSE 0.87 N and steady force with RMSE 0.41 N. Spike timing lag differences did not exceed 6 ms after accounting for handle compliance in the bench setup. Energy dissipated through friction integrated over the retraction exhibited a median difference of 6.5% from simulation to test. These results inform the output comparison scores reported later for each model–mechanism pair.

Output comparison — gradation.

- a. Qualitative trend check on a single specimen without replication.
- b. Quantitative comparison on a limited set with basic error metrics and no replication.
- c. Replicated, multi-condition comparison with predefined metrics and uncertainty; limited blind tests.
- d. Extensive, blinded validation with cross-validation and full error propagation; pre-registered acceptance criteria met.

12. Relevance of validation activities to the CoU (clear): mapping of tests to intended decisions with 0–12 scoring

The validation tests were designed backward from the CoU decisions: to select sheath materials and liners, and to set bench conditions to probe safe envelopes for advancement and release. For crossing, the phantoms span inner diameters and stenosis fractions common to femoropopliteal interventions and embed partial circumferential calcification to replicate asymmetric loading at the lesion shoulder. The imposed advancement speeds (5–20 mm/s) align with motion inferred from fluoroscopy in preclinical studies. For release, the retraction speeds and handle constraints match clinical manipulation ranges. The same fixtures used in simulations were applied in bench tests: distal clamps with measured axial stiffness and proximal actuator compliance were matched to within 10%. Outputs chosen for validation (peak crossing force, force–displacement shape, retraction spike amplitude and timing, steady force) are the precise quantities the models will inform in design and bench planning.

Coverage gaps are acknowledged: tortuous geometries, long calcified segments (>25 mm), and heavily eccentric lesions (>180° arcs) are not included in this assessment and are out of scope for the declared CoU. Fluid-structure interactions are simplified to lubrication through friction coefficients rather than full fluid modeling; this simplification was justified by sensitivity analyses that showed limited influence on axial force outputs under current conditions. Given the alignment on kinematics, fixtures, material ranges, and outputs, the tests provide an adequate foundation for decision-making within the CoU.

Relevance of validation activities to the COU — gradation.

- a. Tests unrelated to the COU or use decisions; no mapping of metrics.
- b. Partial mapping with gaps in key inputs or outputs; qualitative relevance.
- c. Direct mapping of most inputs and outputs; minor omissions justified; quantitative coverage summary.

d.

End-to-end mapping with explicit coverage of uncertainties and extrapolation limits; decision thresholds tied to validation metrics.

13. Software quality assurance (clear): configuration control, automated checks, and 0–12 scoring

The modeling pipeline is managed under Git with a trunk-based workflow and protected branches. All solver decks, pre-processing scripts, and post-processing notebooks are versioned and associated with immutable, checksum-verified input datasets. Containerized environments pin solver versions, libraries, and compilers to ensure reproducibility across machines. Pre-commit hooks enforce style and schema compliance for input decks; continuous integration runs unit tests for mesh generation, contact pairing, and IO consistency with 96% line coverage. Execution provenance includes hardware fingerprints and runtime logs; each simulation stores a manifest with hashes for input and output files to prevent accidental drift. Releases are created for each validation cycle, and all results in this paper use Release 1.7.3 of the modeling toolkit.

Defects are tracked in a public issue tracker with severity and resolution time stamps. Over the last year, five medium-severity issues were resolved (two related to contact pair duplication in complex assemblies; three to regression scripting). No open defects affect the results reported here. Static analyzers and deck linters flagged potential unit mismatches and missing boundary conditions, all of which were addressed before execution. Archiving policies require that raw and processed data remain unchanged; any reprocessing must be performed via new pipelines under a distinct release tag.

$$CI_{\{\text{score}\}} = \frac{\sum_{i=1}^n w_i x_i}{\sum_{i=1}^n w_i} \quad (1)$$

$$\backslash \text{varepsilon}_{\{\text{rel}\}} = \frac{\sum_{i=1}^n w_i x_i}{\sum_{i=1}^n w_i} \quad (2)$$

$$U_{\{\text{val}\}} = \frac{\sum_{i=1}^n w_i x_i}{\sum_{i=1}^n w_i} \quad (3)$$

Software quality assurance — gradation.

- a. Ad hoc file management; undocumented environments; no testing.
- b. Basic version control; limited testing; partial documentation of environments.
- c. Formal change control; automated testing with coverage metrics; containerized, documented environments.
- d. Comprehensive SQA with continuous integration; traceable artifacts; auditable provenance; defect management with documented resolutions.

14. Test samples (clear): sample selection, conditioning, fixtures, and 0–12 scoring

Phantom construction followed a controlled protocol. Inner lumen diameters (5.5–6.5 mm) and wall thickness (1.0 ± 0.05 mm) were achieved by casting silicone (Shore A 40) around precision mandrels and verified by CT (voxel 60 μ m) and optical metrology (± 0.01 mm). Lesions were formed with inserts to achieve 40–80% area reduction; eccentricity was set by partial ring molds at predefined arcs. Calcification heterogeneity was introduced via microbead fillers at 0%, 10%, and 20% volume fraction in a polyurethane matrix injected into the lesion arc; CT quantified the bead distribution, with a coefficient of variation under 8% across samples. The phantoms were hydrated and stored at 37 °C for 24 h before testing. Each specimen underwent 20 cycles of preconditioning at 5 mm/s before recorded trials.

Fixtures matched those represented in the models: distal clamps with axial stiffness 45–60 N/mm and rotational freedom about the vessel axis; proximal actuators with compliance 0.03–0.05 mm/N. Alignment was measured by dial indicators and laser trackers to within 0.2° angular misalignment and 0.1 mm translational offset. The test fluid was 0.9% saline at 37 ± 0.5 °C; temperature and humidity were monitored ($96 \pm 2\%$ RH) throughout. The load cell was calibrated within one month of testing; drift was under 0.05 N over 10 min. All test metadata, including device lot numbers and phantom IDs, were recorded and linked to the corresponding simulation inputs. Together, these practices reduce experimental uncertainty and support meaningful comparison to model outputs.

Test samples — gradation.

- a. Convenience samples without metrology or conditioning controls.
- b. Targeted samples with basic metrology and environment control; fixtures characterized coarsely.
- c. Well-characterized, conditioned samples with fixture compliance and alignment measured; environment controlled.
- d. Statistically representative samples with full metrology and uncertainty budgets; fixtures and environment uncertainties quantified and propagated.

15. Use error (brief): operator technique notes and qualitative considerations

Operator technique influences both maneuvers. For crossing, nuanced wrist motions, micro-advances, and subtle torqueing may alter how the sheath edge negotiates the lesion shoulder, and brief pauses can change stick–slip patterns. For release, varying stabilization of the inner shaft and the cadence of retraction can affect spike characteristics. Our bench and modeling protocols impose controlled, idealized actuation to improve repeatability; however, clinical variability around these maneuvers may lead

to force patterns outside the ranges studied here. We note that additional human factors studies would complement the present assessment by documenting technique distributions and their impact on the outputs of interest.

Use error — gradation.

- a. No consideration of operator technique or variability.
- b. Qualitative description of technique with anecdotal ranges.
- c. Quantitative characterization of technique distributions; limited incorporation in models or tests.
- d. Comprehensive human factors analysis with technique distributions propagated through models and tests.

16. Credibility assessment summary — Model 1

We summarize the factor gradations for Model 1 under both mechanisms using the private 0–12 scale. The basis for each entry is drawn from the evidence presented throughout the paper and condensed here for readability. The acceptance threshold for each row is 8; rows at or above this threshold are considered adequate for decision support within the stated CoU.

17. Credibility assessment summary — Model 2

We summarize the factor gradations for Model 2 under both mechanisms using the same 0–12 scale. The acceptance threshold of 8 applies to each factor $\times$ mechanism row independently.

18. Pairwise results: Model 1 $\times$ Mechanism A — factor scores (0–12) and accept/reject

Under Mechanism A, Model 1 is the intended analysis for estimating the force needed to traverse stenoses. The model reproduced the key bench metrics: peak advancement force averaged 7.1 ± 0.7 N vs 7.4 ± 0.6 N in tests across 12 phantoms, with RMSE 0.58 N and R² 0.94 for force–displacement curves after time alignment. Temporal discretization and contact regularization studies showed peak force changed by 2.7% between 0.25 and 0.1 μ s; kinetic-to-internal energy ratios remained below 0.07 at peak, and added mass stayed below 0.08 of the physical mass in the domains of interest. Verification benchmarks for dynamics and contact were within 1.3–1.8% of reference solutions, and continuous integration reported no tolerance breaches over 18 months. The modeling software pipeline is under strict configuration control with 96% test coverage and immutable datasets. Test samples were dimensionally verified, environmentally controlled, and exercised across stenosis and calcification ranges relevant to the CoU. Scores assigned on the 0–12 scale for this pair were: 11/12 (numerical code verification), 10/12 (numerical solver error), 9/12 (output comparison), 10/12 (relevance), 12/12 (software

QA), and 10/12 (test samples). All exceed the acceptance threshold of 8; the pair is accepted for use within the CoU.

Output comparison — gradation.

- a. A single, non-replicated test used for a spot check; no error bars.
- b. A small set of targeted tests with basic error metrics; limited variability reported.
- c. Replicated tests across multiple conditions with RMSE, bias, and coverage metrics; minor gaps remain.
- d. Blinded or cross-validated datasets with uncertainty propagation; pre-specified acceptance and success within margins.

19. Pairwise results: Model 1 × Mechanism B — factor scores (0–12) and accept/reject

Under Mechanism B, Model 1 is not formulated to represent stent–liner detachment phenomena that dominate retraction spikes. Although the underlying numerical kernels are verified and solver error is controlled, the alignment to bench retraction tests and the direct quantitative comparison of spikes are lacking for this pairing. The bench retraction matrix was designed around Model 2’s capabilities and does not offer a clear mapping to Model 1 outputs. Consequently, scores for output alignment and the connection between validation and intended use are below the threshold. The other factors—numerical cores and software discipline—remain strong but are insufficient to compensate for the lack of pertinent validation. The assigned scores are 11/12 (numerical code verification), 10/12 (numerical solver error), 3/12 (output comparison), 5/12 (relevance), 12/12 (software QA), and 4/12 (test samples). This pair is not accepted for use.

Relevance of validation activities to the COU — gradation.

- a. Tests do not correspond to the model’s intended outputs for the mechanism.
- b. Partial overlap between test conditions and model scope; major outputs unmapped.
- c. Strong overlap with minor omissions; outputs well mapped; residual differences justified.
- d. Full end-to-end replication of the mechanism with direct output mapping and uncertainty coverage.

20. Pairwise results: Model 2 × Mechanism A — factor scores (0–12) and accept/reject

Under Mechanism A, Model 2 focuses on stent–liner interaction during release, not the sheath–lesion mechanics that

dominate crossing. While numerical fidelity and software rigor are high, the output mapping to crossing force–displacement and peak advancement force is not established, and targeted quantitative comparison is absent. Crossing phantoms do not exercise the interactions emphasized in Model 2. The assigned scores are 11/12 (numerical code verification), 9/12 (numerical solver error), 4/12 (output comparison), 5/12 (relevance), 12/12 (software QA), and 4/12 (test samples). This pair does not meet the threshold and is not accepted for use.

Test samples — gradation.

- a. Specimens not representative of mechanism; unknown conditioning.
- b. Some representative features; limited metrology and control.
- c. Representative features with controlled metrology and environment; fixtures characterized.
- d. Mechanism-specific specimens with statistical coverage and propagated uncertainties.

21. Pairwise results: Model 2 × Mechanism B — factor scores (0–12) and accept/reject

Under Mechanism B, Model 2 is the intended analysis for predicting retraction spikes and steady force during stent release. The model reproduced spike amplitudes (12.6 ± 1.3 N vs 13.2 ± 1.0 N) with RMSE 0.87 N and steady force with RMSE 0.41 N across 10 tests. Spike timing lag was within 6 ms of measured values after accounting for handle compliance. Temporal refinement reduced changes in spike amplitude to 3.9% between the last two steps, and spike timing stabilized within 1.9 ms. Mass scaling remained under 0.08 for added-to-physical mass in the critical domains. The modeling kernels passed verification benchmarks, and the software pipeline ensured reproducibility and traceability. The retraction tests were designed to replicate the clinical maneuver and outputs. The assigned scores are 11/12 (numerical code verification), 9/12 (numerical solver error), 10/12 (output comparison), 10/12 (relevance), 12/12 (software QA), and 9/12 (test samples). All exceed the threshold; the pair is accepted for use within the CoU.

Software quality assurance — gradation.

- a. Uncontrolled scripts and data; no testing.
- b. Version control and some tests; limited reproducibility.
- c. Controlled releases, containerized environments, and automated tests with coverage.
- d. Full provenance, audit trails, automated validations, and immutability of artifacts.

22. Synthesis across factors and mechanisms: decision to use within the CoU

Across mechanisms, several consistent threads emerge. Independent mechanics checks reproduced classic impact and contact problems within 1.5% error and showed no regression drift over 18 months, supporting trust in the underlying numerical kernels before model-specific detail is considered. Temporal refinement and controlled contact regularization brought the key force outputs to within 3% (crossing) and 4% (retraction) changes between the final two refinement steps; kinetic energy fractions remained under 7% at peaks, and capping mass augmentation to below 8% avoided inertia artifacts. Agreement with bench was strong in the matched pairs: crossing peak forces of 7.4 ± 0.6 N measured compared to 7.1 ± 0.7 N predicted (RMSE 0.58 N; R2 0.94), and retraction spike amplitudes of 13.2 ± 1.0 N measured compared to 12.6 ± 1.3 N predicted (RMSE 0.87 N; timing lag <6 ms). The experiments used for judging the models mirrored the maneuver sequences and parameter ranges in scope; for example, 82% of lesion morphologies observed in our internal preclinical imaging fall within the phantom parameter ranges, and the actuation speeds reflect clinical practice for straight to mildly curved segments.

The toolchain that produces, executes, and curates the models is tightly controlled: a change-managed environment with containerized solvers, 96% unit test coverage over pre- and post-processing, immutable datasets with checksums, and no unresolved defects on the release used here. Finally, the physical samples and fixtures were prepared and verified with attention to factors that influence force measurement: twelve phantoms with CT-verified lesion geometry and controlled calcification, cyclic preconditioning to reduce transient softening, salt solution at 37 ± 0.5 °C, high-resolution load cells, and fixture alignment within 0.2°.

Given this synthesis, the decision is to use Model 1 for Mechanism A and Model 2 for Mechanism B within the CoU to inform design tradeoffs and bench test planning. The mismatched pairs are not used for decision support because the evidence does not establish a direct mapping between model outputs and the retraction or crossing phenomena, respectively, despite excellent general numerical and software discipline.

Numerical code verification — gradation.

- a. Minimal checks without quantitative thresholds.
- b. A few benchmarks with qualitative expectations.
- c. Benchmarks with numeric tolerances and regression monitoring.
- d. Extensive verification suite with tight tolerances and continuous integration over time.

23. Discussion

The primary outcomes—a pass for Model 1 under lesion crossing and for Model 2 under sheath retraction—reflect a

principled alignment of validation to the CoU. The findings also delineate where confidence is not warranted: applying Model 1 to retraction spikes or Model 2 to crossing forces is not presently justified. This partitioning is instructive in multi-physics device modeling, where the urge to stretch models beyond their intended mechanisms is strong. The discipline of scoring each factor separately for each pair prevents the dilution of such gaps.

Some modeling choices merit reflection. First, the decision to omit full fluid–structure interaction was grounded in sensitivity analyses that showed axial force outputs were only weakly affected by lumen fluid at the velocities in scope, provided lubrication was reflected through appropriate friction coefficients. While this holds for near-straight geometries, higher curvatures or pulsatile conditions may necessitate coupling to a fluid solver or enhanced lubrication models. Second, contact modeling in explicit dynamics is prone to numerical chatter; we mitigated this through moderate contact damping and tight penalty settings, while demonstrating bounded influence on outputs. The convergence of spike timing and amplitude with time-step refinement further supports stability within the selected settings.

The experimental platform aligned well with simulation boundary conditions and kinematics, an alignment that was essential to the strong agreement observed for the matched pairs. The use of CT to verify lesion geometry and calcification distribution increased confidence that contact configurations were as intended. Nonetheless, silicone phantoms cannot reproduce all aspects of arterial wall anisotropy, viscoelasticity, and plaque heterogeneity. Comparisons to ex vivo arteries, while more variable, could further challenge the models and will be valuable as a next step. The present results should therefore be interpreted as sufficient for the narrow CoU but not as general validation across all peripheral vascular anatomies and pathologies.

Regarding the scale and thresholds, the private 0–12 gradation provided granularity to reflect differing strengths among factors. High scores in software discipline and numerical core checks are necessary but not sufficient; they must be paired with output agreement and relevant test design, as underscored by the rejected pairs. In practice, the acceptance threshold of 8 provided a clear demarcation; scenarios with strong verification and solver control but weak validation relevance rightly did not cross the bar.

Finally, forces are only part of the safety story. Local wall strains during crossing and transient contact pressures during release can contribute to endothelial injury or plaque disruption. Model 1 computed local vessel wall strains that, in our simulations, peaked at 8–12% in the lesion shoulder under typical conditions; while we did not validate these strain fields directly in this study, they are plausible in magnitude when compared to indentation tests on silicone plaques. Incorporating direct measurements of local strains or surrogate measures (e.g., optical markers on phantom walls) would enhance confidence in these secondary outputs.

Numerical solver error — gradation.

- a. No exploration of solver-induced error.
- b. Limited sensitivity checks with incomplete reporting.
- c. Systematic exploration with quantitative changes and bounded mass scaling.
- d. Convergence demonstrated with extrapolated estimates and uncertainty bounds carried to conclusions.

24. Limitations and future work

Several limitations constrain the generality of this assessment. Anatomical variability in curvature, lesion length, and eccentricity was restricted: curvature radii were ≥ 60 mm, lesion lengths 15–25 mm, and eccentric arcs 60–120°. Heavily calcified, long, or highly eccentric lesions may alter contact regimes and frictional behavior. Vessel wall material models, while grounded in literature and component tests, used a simplified hyperelastic form without explicit viscoelasticity; time-dependent effects could modulate spike attenuation or crossing hysteresis at lower speeds. The nitinol stent model used a calibrated superelastic law, but manufacturing variability in crimp profiles or surface finish could affect stick–slip beyond the tested lots. Model 1’s strain predictions were not validated directly; a study using digital image correlation on transparent phantoms could provide such data.

On the numerical side, explicit dynamics delivers computational efficiency for short, transient events but imposes small time steps. While we demonstrated bounded solver-induced error at selected steps, full space–time convergence with extrapolation was not performed due to computational cost. Alternative integration schemes or adaptive stepping could be explored to expand operating ranges without compromising stability. Contact algorithms, particularly penalty methods, involve tuning; although our sensitivity studies suggest limited influence within bands, more formal parameter identification against dedicated contact experiments would strengthen confidence.

Future work will extend validation along three axes: (1) anatomy, by introducing controlled curvature (radius 30–60 mm) and measuring its effect on both crossing and retraction; (2) material heterogeneity, by expanding calcification distributions and including fibrotic inclusions with higher anisotropy; and (3) probabilistic uncertainty, by quantifying how input distributions (friction, geometry, compliance) propagate to outputs using non-intrusive polynomial chaos or stochastic collocation. We also plan to instrument phantoms with embedded pressure sensors and surface markers to validate local contact pressures and wall strains, and to append blind prediction phases where model parameters are frozen before new tests are executed.

Regarding human factors, systematic characterization of operator pull and push cadences and stabilization strategies could be integrated into the test protocols, enabling a broader CoU that includes technique variability. Finally, we will evaluate whether simplified surrogate models trained on the FEA outputs can aid rapid design exploration while staying within validated ranges.

Output comparison — gradation.

- a. Ad hoc matches on a subset of outputs; no plan to address gaps.
- b. Targeted tests cover main outputs; gaps noted but not yet addressed.
- c. Comprehensive tests for primary outputs; secondary outputs planned for future validation.
- d. Full validation plan includes primary and secondary outputs with staged, blinded predictions.

25. Conclusions

This study applied a structured, risk-informed credibility assessment to two explicit FEA models for a transcatheter delivery system. The assessment, using a private 0–12 scale with an acceptance threshold of 8 for each model $\times$ mechanism pair, yielded two accepted pairings: Model 1 for lesion crossing and Model 2 for sheath retraction. These pairings demonstrated strong agreement with targeted bench tests, bounded solver-induced errors under systematic refinement, solid alignment of validation activities to the CoU, and rigorous software discipline and test sample control. The cross-mechanism pairings did not meet the threshold due to insufficient validation relevance and output mapping, underscoring the importance of aligning models with the phenomena they are meant to represent.

Within the declared CoU—straight or mildly curved vessels, 40–80% stenoses with partial calcification, and clinically representative actuation speeds—the accepted pairings can inform design down-selection and bench test planning for safe deployment force envelopes and spike management. The approach and evidence presented provide a template for extending credibility to additional mechanisms and anatomies through targeted experiments, deliberate numerical studies, and disciplined software practices.

Relevance of validation activities to the COU — gradation.

- a. Vague linkage between validation and intended use.
- b. Partial mapping with untested critical aspects.
- c. Clear mapping for primary decisions with minor omissions.
- d. Complete mapping for all decisions with quantified uncertainty bounds.

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Table 1: Credibility factor gradations for Model 1 across Mechanism A (lesion crossing) and Mechanism B (sheath retraction).

Credibility factor	Level	Basis
Numerical code verification - Model 1 - Mechanism A	11/12	Taylor bar error 1.3%; Hertzian patch within 1.4%; 18 months of nightly regression without tolerance breaches.
Numerical solver error - Model 1 - Mechanism A	10/12	Peak force changed 2.7% from 0.25 to 0.1 μ s; kinetic/internal energy ratio <0.07; mass scaling <0.08.
Output comparison - Model 1 - Mechanism A	9/12	Peak 7.1 ± 0.7 N vs 7.4 ± 0.6 N; RMSE 0.58 N; R2 0.94 for force-displacement across 12 phantoms.
Relevance of validation activities to the COU - Model 1 - Mechanism A	10/12	Advancement speeds, fixtures, lesion morphologies, and outputs matched to within measured tolerances; gaps limited to tortuosity.
Software quality assurance - Model 1 - Mechanism A	12/12	Release 1.7.3; 96% test coverage; containerized runs; immutable, checksummed datasets; no open defects affecting results.
Test samples - Model 1 - Mechanism A	10/12	CT-verified lesions ($\pm 2.3\%$ diameter); 0–20% calcification; 37 ± 0.5 °C; fixtures aligned within 0.2°; calibrated 50 N load cell.
Numerical code verification - Model 1 - Mechanism B	11/12	Same verified kernels apply; Taylor bar and contact tests within 1.3–1.8%; stable regression history.
Numerical solver error - Model 1 - Mechanism B	10/12	Time step refinement yielded $\leq 3\%$ force change; energy ratios acceptable; mass scaling bounded <0.08.
Output comparison - Model 1 - Mechanism B	3/12	No direct spike validation for this model; only qualitative checks; no RMSE reported for retraction outputs.
Relevance of validation activities to the COU - Model 1 - Mechanism B	5/12	Bench retraction tests exist but not mapped to this model’s outputs; key contact interface (stent–liner) absent.
Software quality assurance - Model 1 - Mechanism B	12/12	Same SQA controls as Mechanism A; reproducible pipelines and audits documented.
Test samples - Model 1 - Mechanism B	4/12	Sheath retraction tests performed but not designed around Model 1’s capabilities; limited interpretability for this pairing.

Table 2: Credibility factor gradations for Model 2 across Mechanism A (lesion crossing) and Mechanism B (sheath retraction).

Credibility factor	Level	Basis
Numerical code verification - Model 2 - Mechanism A	11/12	Taylor bar 1.3%; contact benchmarks within 1.4–1.8%; regression stable across releases.
Numerical solver error - Model 2 - Mechanism A	9/12	Force outputs changed $\leq 4\%$ across 0.25 to 0.1 μ s; mass scaling <0.08; contact damping 0.05–0.2 influenced results by $\leq 2.6\%$.
Output comparison - Model 2 - Mechanism A	4/12	No targeted crossing-force validation for this model; limited observations only; no quantitative RMSE available.
Relevance of validation activities to the COU - Model 2 - Mechanism A	5/12	Crossing tests not tailored to stent–liner interactions central to this model; outputs misaligned with intended crossing metrics.
Software quality assurance - Model 2 - Mechanism A	12/12	Same toolchain controls: containerization, 96% coverage, checksummed datasets, no open issues impacting results.
Test samples - Model 2 - Mechanism A	4/12	Crossing phantoms exist but do not exercise stent–liner contact modeled here; limited sample utility for this pairing.
Numerical code verification - Model 2 - Mechanism B	11/12	Verified dynamics and contact kernels; ring-on-rod friction within 1.8%; regression consistent over 18 months.
Numerical solver error - Model 2 - Mechanism B	9/12	Spike amplitude changed 3.9% across final time-step refinement; spike timing stabilized within 1.9 ms; mass scaling <0.08.
Output comparison - Model 2 - Mechanism B	10/12	Spike 12.6 ± 1.3 N vs 13.2 ± 1.0 N; RMSE 0.87 N; lag <6 ms; steady force RMSE 0.41 N across 10 tests.
Relevance of validation activities to the COU - Model 2 - Mechanism B	10/12	Retraction kinematics, handle compliance, and outputs directly matched; remaining gap is tortuosity not in scope.
Software quality assurance - Model 2 - Mechanism B	12/12	As above: controlled environment, CI, provenance; release 1.7.3 used for all runs.
Test samples - Model 2 - Mechanism B	9/12	Instrumented retraction tests with calibrated fixtures; 3 device lots; environment 37 ± 0.5 °C; high-speed video for timing.